Benders Decomposition with Partial Non-Anticipativity Relaxation for Multi-Stage Stochastic Clean Energy Transition Planning[1]

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Overview

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2. Optimization Model
3. Methodology
4. Computational Study
5. Conclusion

Scope and Contributions

Clean energy transition planning for campus-scale integrated electricity–heat systems under strategic and operational uncertainty

Renewable energy technologies whose costs and efficiencies evolve stochastically across stages

Robust reformulation based on box uncertainty sets for short-term operational uncertainty

Benders decomposition with partial non-anticipativity relaxation

Case study: Middle East Technical University (METU) campus, Ankara

Literature Review

			Generation		Storage		Temporal resolution		
Reference	**Scope**	**Opt. type**	**Cost**	**Eff.**	**Cost**	**Eff.**	**Investment**	**Operation**	**Horizon**
Husein et al. (2018)	SEP	Heuristic	D	–	D	–	Single stage	Hourly	25 years
Prina et al. (2019)	SEP	Heuristic	D	–	D	–	5-year	Hourly	35 years
Gils et al. (2017)	SEP	D MILP	D	–	D	–	Single stage	Hourly	1 year
Moret et al. (2020)	SEP	RO	D	–	D	–	Single stage	Monthly	1 year
Cano et al. (2016)	SEP	MSSP	S	–	S	–	Annual	Hourly	16 years
Guevara et al. (2022)	SEP	MSSP	D	–	D	–	10-year	Monthly	30 years
Heuberger et al. (2017)	SEP*	D MILP	D	–	D	–	Annual	Rep. days	35 years
Powell et al. (2012)	SEP*	ADP	D	–	D	–	Annual	Hourly	20 years
Ioannou et al. (2019)	SEP*	MSSP	S	–	–	–	Annual	Annual	15 years
Rathi & Zhang (2022)	SEP*	MSSP	S	–	D	–	5-year	Rep. day	40 years
Child et al. (2019)	CET	D MILP	D	–	D	–	5-year	Hourly	35 years
Mavromatidis & Petkov (2021)	CET	D MILP	D	D	D	D	5-year	Rep. days	30 years
Tian et al. (2022)	CET	D MILP	D	–	D	–	Annual	Monthly	15 years
Zhao & You (2021)	CET	MSARO	D	–	D	–	Annual	Annual	20 years
Lei et al. (2021)	CET	MSSP	D	–	D	–	5-year	Rep. days	30 years
This study	**CET**	**MSSP**	**S**	**S**	**S**	**S**	**Annual**	**2-hourly**	**15 years**

SEP: Strategic Energy Planning, **CET:** Clean Energy Transition, **D MILP:** Deterministic Mixed-Integer Linear Programming, **MSSP:** Multi-Stage Stochastic Programming, **RO:** Robust Optimization, **MSARO:** Multi-Stage Adaptive Robust Optimization, **ADP:** Approximate Dynamic Programming, **D:** Deterministic, **S:** Stochastic, **Rep.:** Representative.
* SEP studies that support the clean energy transition through mechanisms such as carbon taxes, without an explicit 100% renewable energy target.

Planning Framework

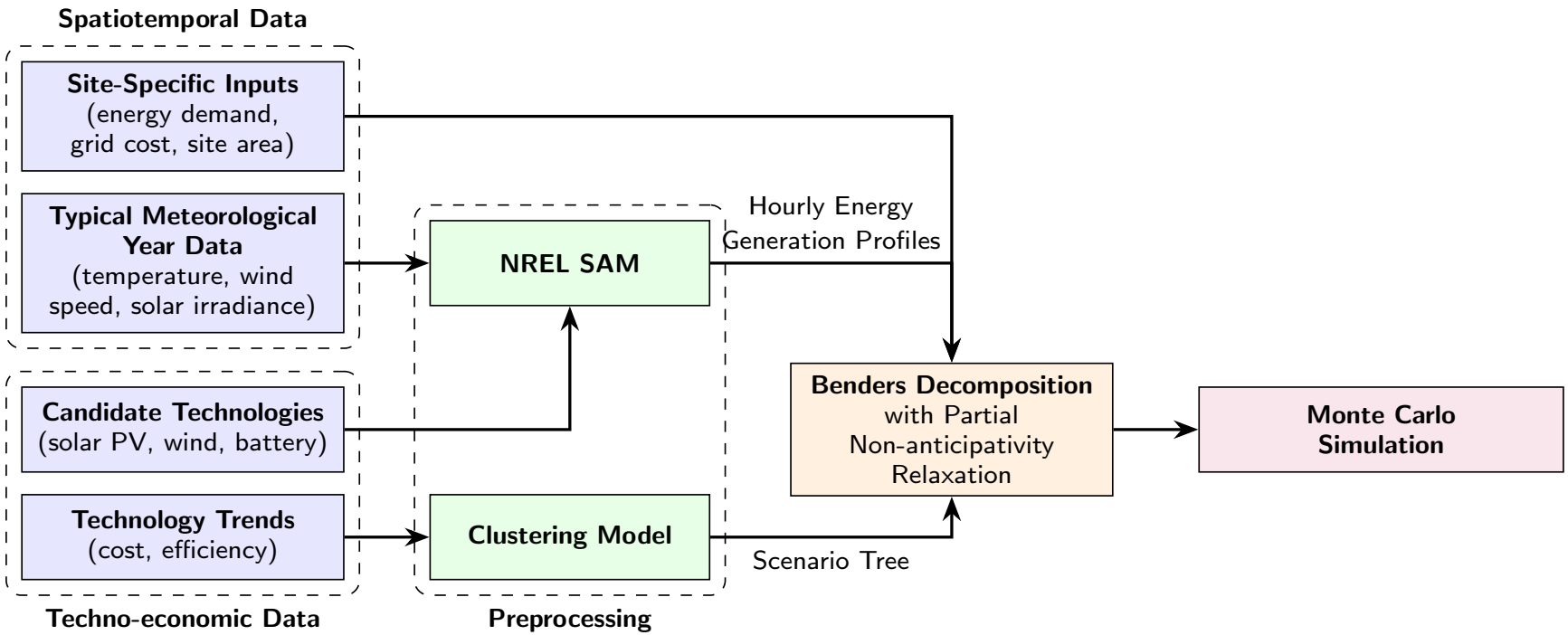


Figure: Overview of the proposed planning framework

Energy System

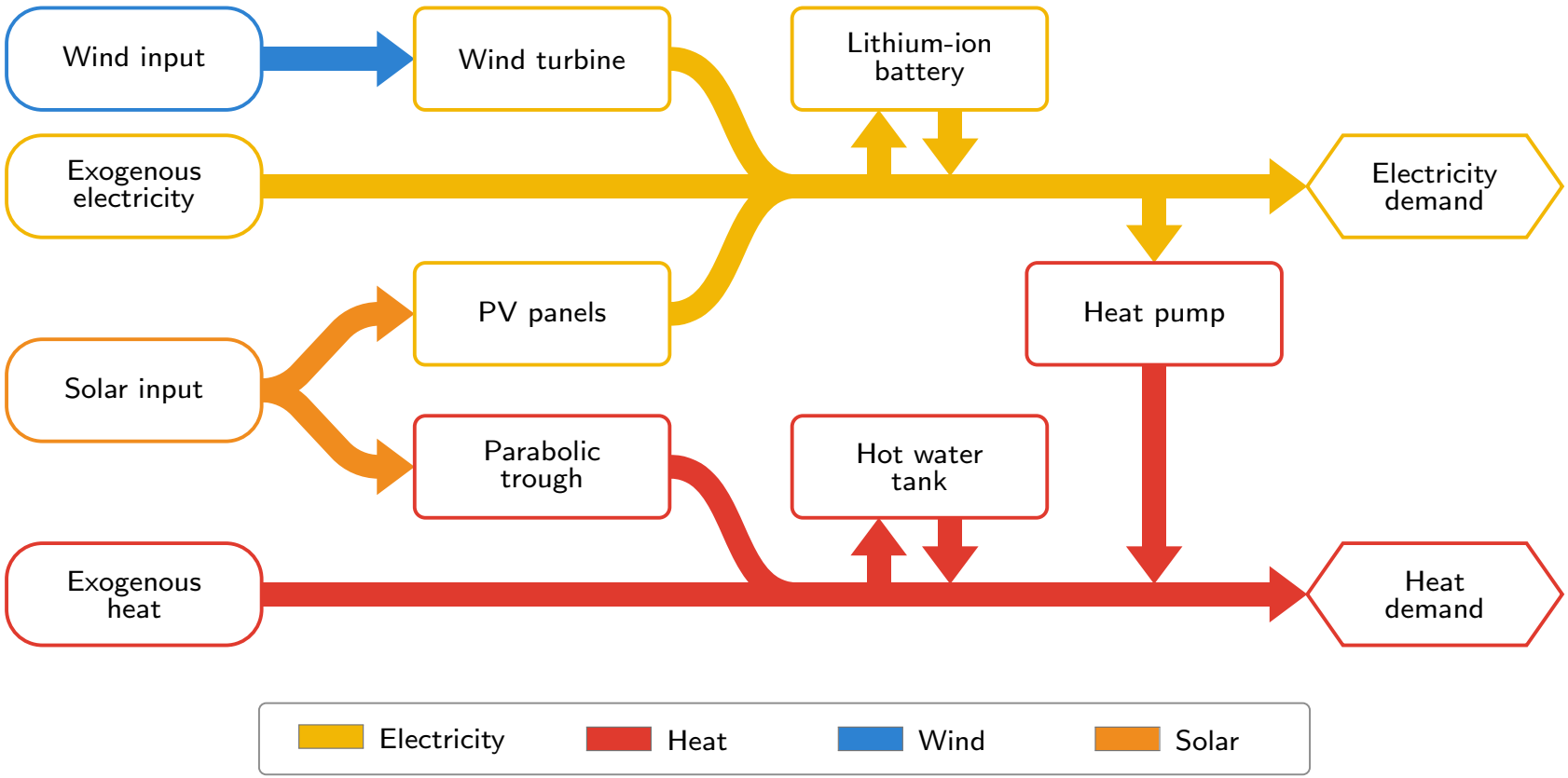


Figure: Schematic of the integrated electricity–heat system.

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Stochastic Model: Decision Variables

$x_{u,t,n}$	Number of installations of technology $u \in \mathcal{U}$ in period $t \in \mathcal{T}_n$ at node $n \in \mathcal{N}$.
$h_{r,q,t,n}$	Energy $r \in \mathcal{R}$ stored at the end of sub-period $q \in \mathcal{Q}$ in period $t \in \mathcal{T}_n$ at node $n \in \mathcal{N}$, carried forward to the next sub-period.
$h^{+}_{r,q,t,n}$	Energy $r \in \mathcal{R}$ charged into storage during sub-period $q \in \mathcal{Q}$ in period $t \in \mathcal{T}_n$ at node $n \in \mathcal{N}$.
$h^{-}_{r,q,t,n}$	Energy $r \in \mathcal{R}$ discharged from storage during sub-period $q \in \mathcal{Q}$ in period $t \in \mathcal{T}_n$ at node $n \in \mathcal{N}$.
$p_{r,q,t,n}$	Energy $r \in \mathcal{R}$ purchased exogenously in sub-period $q \in \mathcal{Q}$ in period $t \in \mathcal{T}_n$ at node $n \in \mathcal{N}$.
$o_{r,q,t,n}$	Energy $r \in \mathcal{R}$ used from inventory and generation in sub-period $q \in \mathcal{Q}$ of period $t \in \mathcal{T}_n$ at node $n \in \mathcal{N}$.
$m_{a,q,t,t',n}$	Heat transferred in sub-period $q \in \mathcal{Q}$ of period $t' \in \mathcal{T}_n$ at node $n \in \mathcal{N}$ by technology $a \in \mathcal{A}$ installed in period $t \in \mathcal{T}'_{a,[t')}$.

Stochastic Model Formulation

$$\text{minimize} \sum_{n \in \mathcal{N} \setminus \{0\}} \sum_{t \in \mathcal{T}_n} \pi_n \beta^{(t-1)} \left(\sum_{u \in \mathcal{U}} (\alpha_{u,n} + \hat{\alpha}_{u,t})\, x_{u,t,n} + \sum_{r \in \mathcal{R}} \sum_{q \in \mathcal{Q}} \lambda_{r,t}\, p_{r,q,t,n} \right)$$

$$o_{r,q,t',n} + \sum_{a \in \mathcal{A}} \sum_{t \in \mathcal{T}'_{a,[t')}} \mathbb{1}_{a,r,q,t,t',n}\, m_{a,q,t,t',n} \geq \delta_{r,q,t'} \qquad \text{(demand 1)}$$

$$p_{r,q,t',n} - h^{+}_{r,q,t',n} + h^{-}_{r,q,t',n} + \sum_{j \in \mathcal{J}} \sum_{t \in \mathcal{T}'_{j,[t')}} \Gamma_{j,r,q,t,t',n}\, x_{j,t,\mu_{t,n}} \geq o_{r,q,t',n} \qquad \text{(demand 2)}$$

$$h_{r,q,t',n} = \eta_r h_{r,\kappa_{q,t',n}} + \eta^{+}_r h^{+}_{r,q,t',n} - h^{-}_{r,q,t',n} / \eta^{-}_r \qquad \text{(storage balance)}$$

$$h_{r,q,t',n} \leq \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}'_{s,[t')}} \upsilon_{s,r,t,t'}\, x_{s,t,\mu_{t,n}} \qquad \text{(storage capacity)}$$

$$m_{a,q,t,t',n} \leq \varphi_a\, x_{a,t,\mu_{t,n}} \qquad \text{(heat transfer capacity)}$$

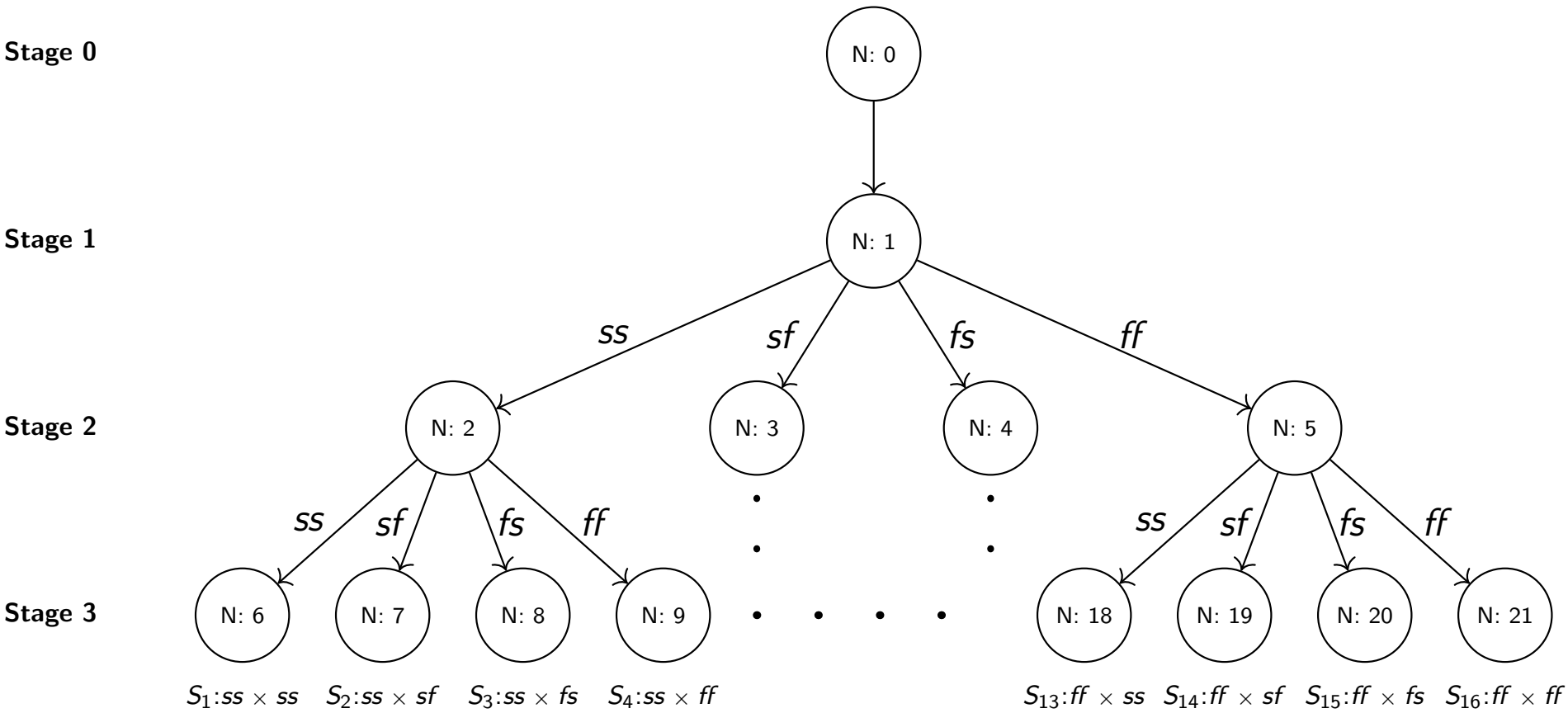
$$\sum_{q \in \mathcal{Q}} p_{r,q,T,n} \leq \gamma_r \sum_{q \in \mathcal{Q}} \delta_{r,q,T} \qquad \text{(emissions)}$$

$$\sum_{u \in \mathcal{U}} \alpha_{u,n}\, x_{u,t,n} \leq \phi_t \qquad \text{(budget)}$$

$$\sum_{u \in \mathcal{U}} \sum_{t \in \mathcal{T}'_{u,[t')}} \rho_{u,t}\, x_{u,t,\mu_{t,n}} \leq \varkappa_{t'} \qquad \text{(space)}$$

$$\left\lfloor \frac{\phi_t}{\alpha_{u,n}} \right\rfloor \geq x_{u,t,n}, \quad x_{u,t,n} \in \mathbb{Z}_+ \qquad p_{r,q,t,n},\ h_{r,q,t,n},\ h^{+}_{r,q,t,n},\ h^{-}_{r,q,t,n},\ o_{r,q,t,n},\ m_{a,q,t,t',n} \in \mathbb{R}_+$$

Three-Stage Scenario Tree



(ss): Slow PV progress, slow battery progress
(sf): Slow PV progress, fast battery progress
(fs): Fast PV progress, slow battery progress
(ff): Fast PV progress, fast battery progress

Derivation of Valid Inequalities

Proposition

For each $r \in \mathcal{R}$, the following inequalities are valid for the MSSP model when $\gamma_r = 0$:

$$\eta_r^{-}\eta_r \sum_{s\in\mathcal{S}} \sum_{t\in\mathcal{T}'_{s,[\hat{\kappa}_{\underline{q},T,n})}} \upsilon_{s,r,t,\hat{\kappa}_{\underline{q},T,n}} x_{s,t,\mu_{t,n}} + \sum_{q=\underline{q}}^{\overline{q}} \sum_{j\in\mathcal{J}} \sum_{t\in\mathcal{T}'_{j,[T)}} \Gamma_{j,r,q,t,T,n} x_{j,t,\mu_{t,n}}$$

$$+ \sum_{q=\underline{q}}^{\overline{q}} \sum_{a\in\mathcal{A}} \sum_{t\in\mathcal{T}'_{a,[T)}} \varsigma_{a,r,t,T} x_{a,t,\mu_{t,n}} \geq \sum_{q=\underline{q}}^{\overline{q}} \delta_{r,q,T} \qquad r \in \mathcal{R}, n \in \mathcal{N}_l, \underline{q} \in \mathcal{Q}, \overline{q} \in \{\underline{q}, \ldots, Q\} \quad (2)$$

where the coefficients $\varsigma_{a,r,t,T}$ are defined as follows:

$$\varsigma_{a,r,t,T} = \begin{cases} \varphi_a\ (1 - \sigma_{a,t,T}), & \text{if } r = \text{heat} \\ 0, & \text{if } r = \text{electricity} \end{cases} \qquad a \in \mathcal{A}, t \in \mathcal{T}'_{a,[T)} \quad (3)$$

$$V_{\omega,r}(x_\omega^{(k)}, \underline{q}, \overline{q}) := \eta_r^{-}\eta_r \sum_{s\in\mathcal{S}} \sum_{t\in\mathcal{T}'_{s,[\hat{\kappa}_{\underline{q},T,n_\omega})}} \upsilon_{s,r,t,\hat{\kappa}_{\underline{q},T,n_\omega}} x^{(k)}_{s,t,\mu_{t,n_\omega}}$$

$$+ \sum_{q=\underline{q}}^{\overline{q}} \sum_{j\in\mathcal{J}} \sum_{t\in\mathcal{T}'_{j,[T)}} \Gamma_{j,r,q,t,T,n_\omega} x^{(k)}_{j,t,\mu_{t,n_\omega}} + \sum_{q=\underline{q}}^{\overline{q}} \sum_{a\in\mathcal{A}} \sum_{t\in\mathcal{T}'_{a,[T)}} \varsigma_{a,r,t,T} x^{(k)}_{a,t,\mu_{t,n_\omega}} - \sum_{q=\underline{q}}^{\overline{q}} \delta_{r,q,T} \quad (4)$$

$$\min_{1\leq\underline{q}\leq\overline{q}\leq Q} V_{\omega,r}(x_\omega^{(k)}, \underline{q}, \overline{q}) \quad (5)$$

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Benders Decomposition with Partial Non-anticipativity Relaxation

Compact model formulation:

$$\min_{x\in\mathcal{X},y\in\mathcal{Y}} \left\{ c^\top x + d^\top y : Ax + By \geq b \right\}. \tag{6}$$

Scenario-wise formulation with non-anticipativity relaxed for y:

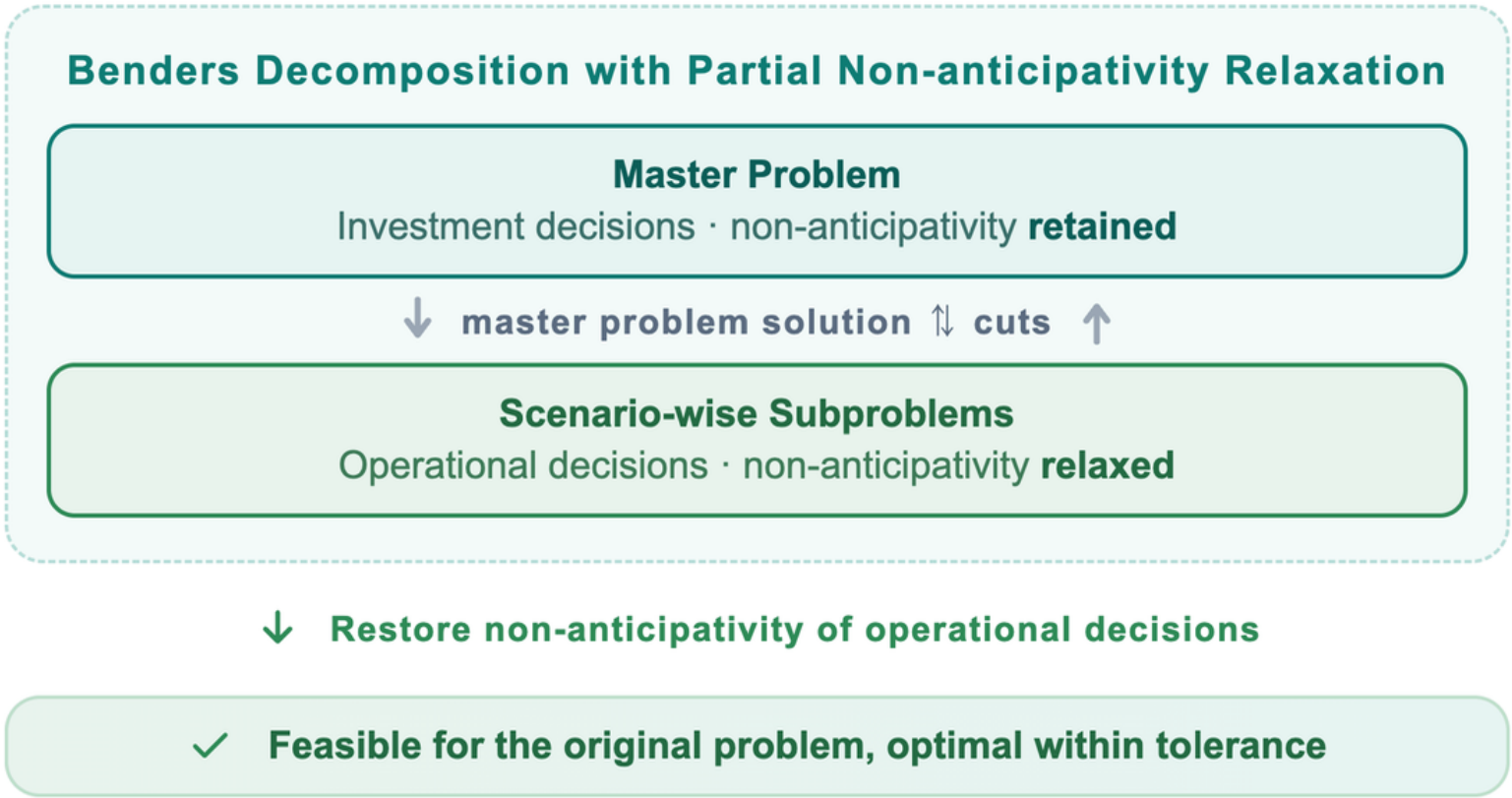
$$\min_{(x_\omega,y_\omega)_{\omega\in\Omega}} \left\{ \sum_{\omega\in\Omega} \Pi_\omega (c_\omega^\top x_\omega + d_\omega^\top y_\omega) : \begin{array}{ll} A_\omega x_\omega + B_\omega y_\omega \geq b_\omega, & \omega\in\Omega \\ y_\omega \in \mathcal{Y}_\omega, & \omega\in\Omega \\ (x_\omega)_{\omega\in\Omega} \in \mathcal{X} & \end{array} \right\}. \tag{7}$$

Benders decomposition with partial non-anticipativity relaxation:

$$\textbf{\textit{MP}} : \min_{(x_\omega,\,\theta_\omega)_{\omega\in\Omega}} \left\{ \sum_{\omega\in\Omega} \Pi_\omega \left(c_\omega^\top x_\omega + \theta_\omega\right) \;:\; \begin{array}{ll} (x_\omega)_{\omega\in\Omega} \in \mathcal{X}, & \\ \theta_\omega \geq (\lambda_\omega)^\top \left(b_\omega - A_\omega x_\omega\right), & \omega\in\Omega,\ \lambda_\omega \in \Lambda_\omega^{\text{opt}} \\ 0 \geq (\mu_\omega)^\top \left(b_\omega - A_\omega x_\omega\right), & \omega\in\Omega,\ \mu_\omega \in \Lambda_\omega^{\text{feas}} \end{array} \right\}. \tag{8}$$

$$\textbf{\textit{SP}}_\omega(x_\omega^*) : \min_{y_\omega\in\mathcal{Y}_\omega} \left\{ d_\omega^\top y_\omega \;:\; B_\omega y_\omega \geq b_\omega - A_\omega x_\omega^* \right\}, \quad \omega\in\Omega. \tag{9}$$

Benders Decomposition with Partial Non-anticipativity Relaxation



Proposition

Let $(x^*_\omega, y^*_\omega)_{\omega\in\Omega}$ be an incumbent of the non-anticipativity-relaxed problem. Assume that each node in the final stage consists of more than one period. Then, there exists a non-anticipative solution $(x^*_\omega, \hat{y}_\omega)_{\omega\in\Omega}$ of the original model whose objective value exceeds that of the incumbent by at most

$$\sum_{r\in\mathcal{R}} \sum_{n\in\mathcal{N}\setminus\mathcal{N}_l} \sum_{n'\in\overline{\mu}_n} \frac{\pi_{n'}\, \beta^{(\underline{t}_{n'}-1)}\, \lambda_{r,\underline{t}_{n'}}}{\eta_r^+} \left(\Delta_{r,n,n'}\right)^+, \tag{10}$$

where $(\cdot)^+ := \max\{0,\cdot\}$ and $\Delta_{r,n,n'}$ is the residual of the inter-node storage-balance constraints, evaluated at the node-wise solution $\tilde{y}$ obtained when these constraints are relaxed:

$$\Delta_{r,n,n'} := \tilde{h}_{r,1,\underline{t}_{n'},n'} - \eta_r\, \tilde{h}_{r,Q,\overline{t}_n,n} - \eta_r^+\, \tilde{h}^+_{r,1,\underline{t}_{n'},n'} + \frac{\tilde{h}^-_{r,1,\underline{t}_{n'},n'}}{\eta_r^-}. \tag{11}$$

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Computational Environment

64-bit Apple Silicon MacBook Pro with an Apple M3 Pro processor (6 performance and 6 efficiency cores) and 36 GB of unified memory

Python 3.13.6

Gurobi 13.0.2

4-hour time limit for each experiment

Base Case

A dynamic strategic plan for the clean energy transition of the METU campus over a 15-year planning horizon (2026–2040).

The planning horizon comprises three stages of five years each.

Each year comprises 4,368 sub-periods of two hours each.

Investment decisions are annual, and operational decisions are made at a two-hour resolution.

The scenario tree contains 16 leaf nodes and 16 scenario paths.

9,435,001 constraints and 10,352,904 decision variables, including 742 investment variables, of which 318 are integer.

Computational Performance

Table: LP relaxation: solution times in seconds and optimality gaps in %.

	Extensive Form		Single-Phase Relaxed Benders				Relaxed Benders				Relaxed Benders + VIs			
	Time	Gap	Mstr.	Sub.	Total	Gap	Mstr.	Sub.	Total	Gap	Mstr.	Sub.	Total	Gap
24-h	107	0.00	2	38	41	0.02	2	39	41	0.02	2	31	34	0.04
8-h	3,450	0.00	17	411	432	0.03	8	312	323	0.56	8	281	292	0.43
4-h	†	–	57	1,588	1,656	0.00	11	911	927	0.71	12	882	901	0.73
2-h	†	–	178	8,084	8,330	0.00	17	3,290	3,320	0.96	23	3,710	3,757	0.89

Table: MILP results: time, optimality gap, and upper bound (UB) in $M.

	Extensive Form		Classical Benders				Relaxed Benders				Relaxed Benders + VIs				
	Time	Gap	Mstr.	Sub.	Total	Gap	Mstr.	Sub.	Total	Gap	Mstr.	Sub.	Total	UB	Gap
24-h	1,063	0.90	1,912	1,897	3,821	1.00	41	44	86	0.45	73	40	115	123.15	0.48
8-h	†	1.84	2,490	11,878	†	4.68	367	394	767	0.84	266	376	650	156.73	0.69
4-h	†	–	914	13,476	†	23.46	385	1,256	1,654	1.00	357	1,255	1,628	158.81	0.95
2-h	†	–	144	14,254	†	–	1,842	4,813	6,699	0.97	1,144	4,760	5,956	162.27	0.99

VIs: valid inequalities. Mstr.: master problem time. Sub.: subproblem time.
† 4-hour time limit reached. Dashes indicate that no optimality gap was available.

Table: Out-of-sample Monte Carlo simulation results for nominal and robust investment plans under white- and pink-noise deviations.

Investment ϵ_i	Noise	Dispatch ϵ_d	$\bar{I} = 1$				
			Purchasing cost ($M)	Purchased (MWh)		Shortfall (%)	
				Electricity	Heat	Electricity	Heat
0	White	0	35.03	88,948	746,917	0.152	0.398
		1	34.76	86,775	745,658	0.006	0.064
		2	35.08	86,497	757,951	0.000	0.078
	Pink	0	35.78	93,550	757,007	1.309	2.363
		1	35.16	88,868	751,813	0.202	0.877
		2	35.38	88,083	762,360	0.037	0.637
0.1	White	0	33.08	78,120	728,475	0.131	0.281
		1	32.76	75,525	727,145	0.004	0.033
		2	33.10	75,259	740,205	0.000	0.035
	Pink	0	33.87	83,172	737,764	1.088	1.967
		1	33.19	77,966	732,764	0.153	0.641
		2	33.43	77,154	744,097	0.025	0.433
0.2	White	0	31.93	73,040	709,055	0.121	0.244
		1	31.65	70,664	708,453	0.004	0.022
		2	32.04	70,314	723,484	0.000	0.017
	Pink	0	32.73	77,890	719,404	1.034	1.767
		1	32.10	73,037	714,874	0.143	0.544
		2	32.39	72,200	728,013	0.021	0.348

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Conclusions and Future Work

A multi-stage stochastic programming framework for clean energy transition planning that captures technological and short-term operational uncertainty at high temporal resolution.

- Uncertain technological progress
- High-resolution operational dynamics
- Short-term uncertainty in demand, renewable generation, and heat pump performance

Benders decomposition with partial non-anticipativity relaxation solves the resulting large-scale problems. A correction LP restores operational non-anticipativity at termination.

Thank you for your attention

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